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HZSM-5 Pelletized and Modified with α-$Ca_3(PO_4)_2$ and HPO_4^{2-} as a Catalyst for Methanol Conversion

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ABSTRACT

Mechanically strong pellets of HZSM-5 were prepared by co-milling with α-$Ca_3(PO_4)_2$, extruding into pellets and hydrothermally treating them in HPO_4^{2-} solution. The catalyst life was significantly longer than that of the parent HZSM-5 in the conversion of methanol to light olefins at high temperatures. The modification decreased catalyst deactivation due to both coking and dealumination. The α-$Ca_3(PO_4)_2$ and non-stoichiometric hydroxyapatite transformed from α-$Ca_3(PO_4)_2$ were considered to reduce the acid sites on the external surface of HZSM-5 crystals and to supply HPO_4^{2-} gradually, resulting in extension of the catalyst life.

INTRODUCTION

HZSM-5-type zeolites have been investigated widely as catalysts for methanol conversion to light olefins [1], and a higher selectivity of light olefins has been reported for HZSM-5 with a larger Si-to-Al ratio [2–7]. Various modifications of HZSM-5 have been recommended in order to achieve better catalyst performance [8–11].

HZSM-5 catalyst for methanol conversion at high temperatures deteriorates rapidly owing to coking and dealumination of the zeolite framework in the presence of by-product water. At high temperatures ($>500°C$) dealumination seems more or less inevitable [12].

Coke generation is caused by strong acid-catalysed oligomerization and cy-

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clization of olefins to fused-ring aromatic compounds. For this reason, various modifications of HZSM-5 with basic compounds have been tried in order to neutralize strong acid sites [13,14].

The relatively fast deposition of coke on the external surface of a zeolite results in poisoning of the acid sites and plugging of the pores, preventing access of the reactants to intracrystalline acid sites. This coke deposition on the external surface leads to rapid deactivation at an early stage of methanol conversion. Therefore selective dealumination [15] or poisoning of the acid sites on the external surface [2,3] is of particular interest.

Crystalline zeolite powders should be shaped into mechanically strong pellets in an economical way for easy handling of industrial catalysts. However, zeolite crystals show poor cohesion, so an inorganic binder such as water-glass, colloidal silica, colloidal alumina or natural clay has to be employed when extruding pellets of zeolite. When methanol conversion is conducted at high temperatures, however, such inorganic binders tend to catalyse conversion to undesirable products (CO, CO_2, CH_4, H_2, etc.). Hence binders for mechanically strong zeolite pellets are required that do not significantly influence the yield and selectivity of light olefins.

We recently found that α-$Ca_3(PO_4)_2$ is an outstanding binder for non-fragile pellets of HZSM-5 microcrystals, and also that the pellets subsequently treated with a phosphorus compound are superior as long-life catalysts for methanol conversion to light olefins at high temperature (600°C). In this paper we describe the manufacture of mechanically strong zeolite pellets and methanol conversion results, and propose an explanation of the role of phosphorus compounds.

EXPERIMENTAL

Preparation of HZSM-5 [7]

A hydrogel consisting of colloidal silica (Shokubai Kasei; 30–31 wt.-% of SiO_2, 0.36–0.47 wt.-% of Na_2O, 0.03 wt.-% of Al_2O_3), analytical-reagent grade aluminium nitrate, sodium hydroxide and tetrapropylammonium bromide (TPABr), with Si-to-Al, OH^--to-Si, H_2O-to-Si and TPABr-to-Si ratios of 400, 0.32, 80 and 0.051, respectively, was stirred in a PTFE -coated stainless-steel reactor and refluxed at 100°C for 8 days.

The precipitated zeolite microcrystals were separated and washed repeatedly by a centrifuge method, dried at 120°C and calcined at 520°C for 16 h. The NaZSM-5 thus obtained was ion exchanged twice with 0.6 *M* hydrochloric acid (10 ml/g zeolite) at room temperature, then the HZSM-5 powder was washed, dried and calcined.

The yield of HZSM-5 was nearly 70%. Its characteristics were as follows: BET surface area, 330 m^2/g; Si-to-Al molar ratio, 250; sodium contents, <0.01

wt.-%. The X-ray diffraction (XRD) pattern of the prepared HZSM-5 was consistent with that reported elsewhere for HZSM-5.

Preparation of calcined phosphates

Calcium phosphate ($Ca_2P_2O_7$) was obtained by the calcination of analytical-reagent grade $CaHPO_4 \cdot 2H_2O$ at 500°C. β-$Ca_3(PO_4)_2$ was prepared by the reaction of $NH_4H_2PO_4$ with $Ca(OAc)_2 \cdot 2H_2O$ at 80°C. α-$Ca_3(PO_4)_2$ was synthesized according to the procedure described by Monma [17]. Hydroxyapatite of analytical-reagent grade and all other calcium phosphates were well characterized by XRD analysis.

Pelletization of HZSM-5 with α-$Ca_3(PO_4)_2$

HZSM-5 (10 g) and α-$Ca_3(PO_4)_2$ (0.5–10 g) were ground together to a fine powder, followed by neading with water and methylcellose. Wet pellets of 3 mm in diameter and 5 mm long were produced by extrusion, dried at 100°C and calcined at 520°C for 6 h. The hardness of the pellets was measured by means of a hardness tester.

Hydrothermal treatment of pellets

A 5-g amount of pellets was immersed in 80 ml of water, a slightly acidic solution [10 mmol of $(NH_4)_2HPO_4$ plus 30 mmol of ammonium acetate) or a basic solution (10 mmole of NH_4OH) at 80°C for 44 h. After washing the pellets with water, they were dried at 120°C and calcined at 520°C for 3 h.

Steaming

A 1.8-g amount of pellets was steamed in a flow of 100% steam (3 g/h) at 600°C.

Adsorption of hexane isomers

A mixture of equal amounts of 2,2-dimethylhexane (effective molecular size 6.2 Å), 3-methylheptane (5.0 Å) and n-hexane (4.3 Å) was introduced as a 2-μl pulse into a gaschromatographic (GC) column containing a sample, previously preheated to 500°C in helium. The adsorption of the samples were characterized as the total numbers of the pulses required until the pulse-supplied hexane isomers were detected in the effluent stream.

X-ray diffraction analysis of calcium phosphates

Samples for XRD analysis were previously calcined at 900 °C for 1.5 h. XRD patterns (Cu K radiation, Ni filter) were recorded on a Philips PW1700 instrument, scanning from 21 to 35 °2θ with an interval of 0.07 °/s.

Methanol conversion

Reactions were conducted in a quartz fixed-bed flow reactor, using 0.45-0.68 g of catalyst of particle size 12–24 mesh at 600 °C. The reaction gas, methanol–argon (1:1), was fed at a WHSV (g methanol/ g HZSM-5 per hour) of 7.0 h^{-1}. The products were analysed by on-line GC on Porapak Q for water, carbon dioxide, dimethyl ether, ethane, ethylene, propane and propylene, on 13X molecular sieve for hydrogen, carbon monoxide, argon and methane, on activated alumina for C_3–C_5 hydrocarbons, and on polyethylene glycol for methanol, ethanol, BTX (benzene, toluene and xylenes) and other $C_{\geqslant 6}$ hydrocarbons.

The yields and selectivities of the products were calculated using argon as an internal standard. The percentage yields given represent the carbon contents of particular hydrocarbons relative to that of total hydrocarbons produced in addition to dimethyl ether.

RESULTS AND DISCUSSION

Transformation of α-$Ca_3(PO_4)_2$

Monma et al. [16,17] reported that in an aqueous solution of pH higher than 5 at about 80 °C, α-$Ca_3(PO_4)_2$ (α-TCP) is hydrolysed to non-stoichiometric hydroxyapatite (NHA), $Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$, where x is in the range 0–1. The hydrolysed product forms a very hard aggregate of cylindrical rods or thin blades, and decomposes thermally to β-$Ca_3(PO_4)_2$ (β-TCP) and a stoichiometric hydroxyapatite at 800 °C according to the following equation:

$$Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x} \rightarrow 3x\beta\text{-}Ca_3(PO_4)_2 + (1-x)Ca_{10}(PO_4)_6(OH)_2$$

XRD patterns for the hydrolysed product after calcination at more than 800 °C show peaks characteristic of β-TCP and hydroxyapatite. The relative peak intensities are proportional to x in the formula shown above.

In a solution of high pH, hydrolysis of α-TCP to NHA proceeds slowly and x is nearly zero (stoichiometric hydroxyapatite formation). On hydrolysis in a solution of low pH (<5), the products are $CaHPO_4 \cdot 2H_2O$ and octacalcium phosphate. The former decomposes to $Ca_2P_2O_7$ and the latter to β-TCP and $Ca_2P_2O_7$ at 500 °C.

We examined the hydrolysis of α-TCP in water at 80 °C and compared by XRD analysis the products after calcination at 900 °C with β-TCP and hy-

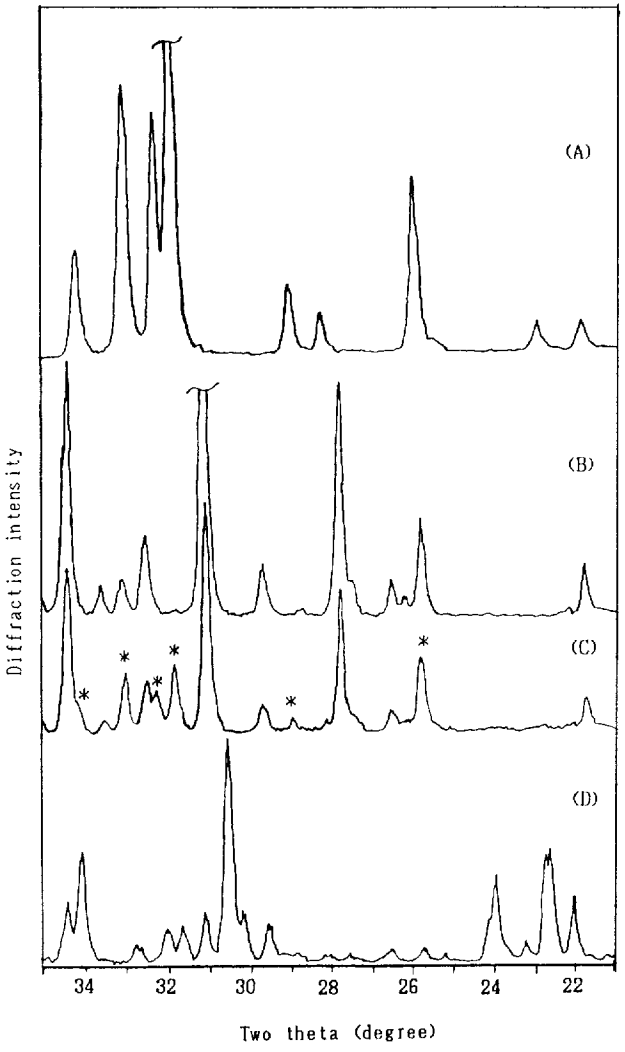

Fig. 1. X-ray diffraction patterns of various calcium phosphate calcined at 900°C: (A) hydroxyapatite; (B) β-TCP; (C) α-TCP/H_2O; (D) α-TCP; (*) hydroxyapatite.

droxyapatite (Fig. 1). Peaks assigned to α-TCP disappeared completely on hydrothermal treatment for 44 h, and intense peaks of β-TCP and weak peaks of hydroxyapatite (marked with asterisks) were observed instead.

Preparation of pellets from HZSM-5 and phosphorus compounds

Table 1 lists the variously modified HZSM-5 pellets produced by extrusion after grinding together HZSM-5 and α-TCP followed by various hydrothermal

TABLE 1

Modified HZSM-5 catalysts for methanol conversion

No.	Modification of HZSM-5 (α-TCP addition%)	Relative Pellet hardness (kg)	XRD analysis of calcium phosphates calcined at 900°C	Catalysis lifetime (h)[a]
1	HZSM-5	<0.1	-----	8
1P	HZSM-5/HPO_4^{2-}	<0.1	-----	21
2	HZSM-5/α-TCP(5)	0.1	[b]	14
2P	HZSM-5/α-TCP(5)/HPO_4^{2-}	0.5	[b]	91
2B	HZSM-5/α-TCP(5)/OH^-	0.4	[b]	32
3	HZSM-5/α-TCP(20)	0.5	α-TCP + β-TCP[c]	30
3P	HZSM-5/α-TCP(20)/HPO_4^{2-}	3.4	β-TCP + hydroxyapatite[c]	84
3B	HZSM-5/α-TCP(20)/OH^-	2.6	Hydroxyapatite	34
4	HZSM-5/α-TCP(30)	1.2	-----	---
4P	HZSM-5/α-TCP(30)/HPO_4^{2-}	6.0	β-TCP + hydroxyapatite[c]	64
5	HZSM-5/α-TCP(34)	2.9	-----	12
%a	HZSM-5/α-TCP(34)H_2O	6.3	β-TCP + hydroxyapatite[c]	26
5B	HZSM-5/α-TCP(34)/OH^-	6.7	β-TCP + hydroxyapatite	64
5P	HZSM-5/α-TCP(34)/HPO_4^{2-}	2.9	β-TCP + hydroxyapatite	63
6	HZSM-5/α-TCP(50)	2.5	-----	---
6P	HZSM-5/α-TCP(50)/HPO_4^{2-}	12	β-TCP + hydroxyapatite[c] + β-$Ca_2P_2O_7$[c]	36

[a]Reaction times to hold the selectivity of $C_2^= + C_3^=$ more than 50%.
[b]Identification of calcium phosphates was impossible because of the low intensities of XRD peaks assigned to the calcium phosphates.
[c]A small amount of calcium phosphate was detected from XRD peaks.

treatments. The catalysts indicated as HZSM-5/α-TCP/HPO_2^{2-} and HZSM-5/α-TCP/HO^- represent the pellets hydrothermally treated in a solution of $(NH_4)_2HPO_4$ and ammonium acetate and in ammonium solution, respectively. The mechanical hardness of the pellets is also given in Table 1.

When up to 5% of α-TCP was added to HZSM-5, the pellet hardness was not much improved with or without subsequent hydrothermal treatment. As the amount of α-TCP added increased (catalysts 3, 4, 5 and 6) the pellets hardened significantly without any hydrothermal treatment. As the process of pelletization includes drying the pellets at 100°C for a short period, hydrolysis of α-TCP in the wet pellets may occur to some extent. A considerable improvement in pellet hardness was achieved by hydrothermal treatment in a slightly acidic solution of $(NH_4)_2HPO_4$ and ammonium acetate (catalysts 3P, 4P and 6P). In order to produce practically strong pellets (relative pellet hardness of more than 2 kg), the addition of more than 10% of α-TCP seems necessary.

Microscopic observation of the pellets

For observation of the morphology and texture of the pellets with α-TCP as binder, scanning electron micrographs of the pellets of catalysts 2 and 6P were

compared, as shown in Fig. 2. The agglomerates of very fine HZSM-5 crystals were observed in Fig. 2a, but calcium phosphate (5%) was not identified. On the other hand, for the pellet of catalyst 6P, tiny thin-bladed (approximately 0.6 μm in length) aggregates were observed among the aggregates of HZSM-5 crystals in Fig. 2b. From the observed difference and the description by Monma

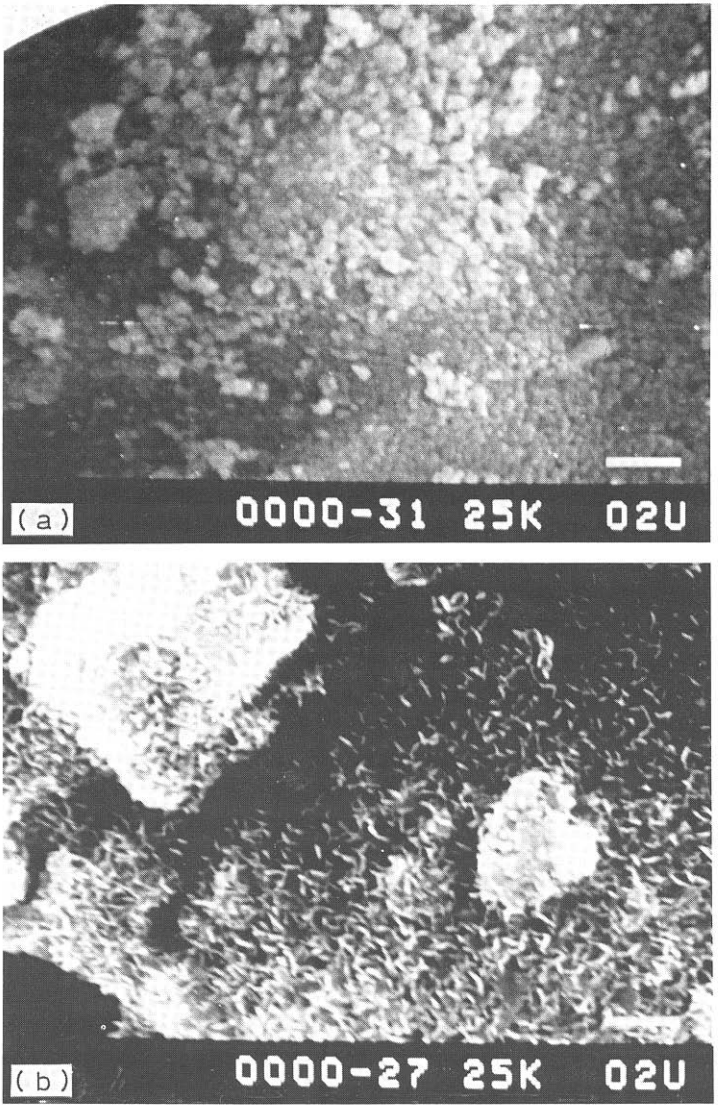

Fig. 2. Scanning electron micrographs of catalyst pellets: (a) HZSM-5/α-TCP(5%); (b) HZSM-5/α-TCP(50%)/HPO_4^{2-}.

et al. [16], it is certain that the pellet hardness is due to the formation of thin blades of non-stoichiometric hydroxyapatite (NHA).

X-ray diffraction analysis of the pellets

In order to identify the conversion of α-TCP to NHA, XRD analysis was carried out on pellets calcined at 900°C. The results are shown in Table 1 and Fig. 3.

Calcination at 900°C of the pellets treated in a slightly acidic solution yields β-TCP as the predominant phase and a small amount of hydroxyapatite (marked with asterisks), as shown in Fig. 3F, G, and H. These results indicate that the α-TCP is converted into NHA, $Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$, where $x \approx 1$. However, on hydrothermal treatment under basic conditions, α-TCP is converted into NHA (Fig. 3J) with $x \approx 0$, i.e., $Ca_{10}(PO_4)_6(OH)_2$, and

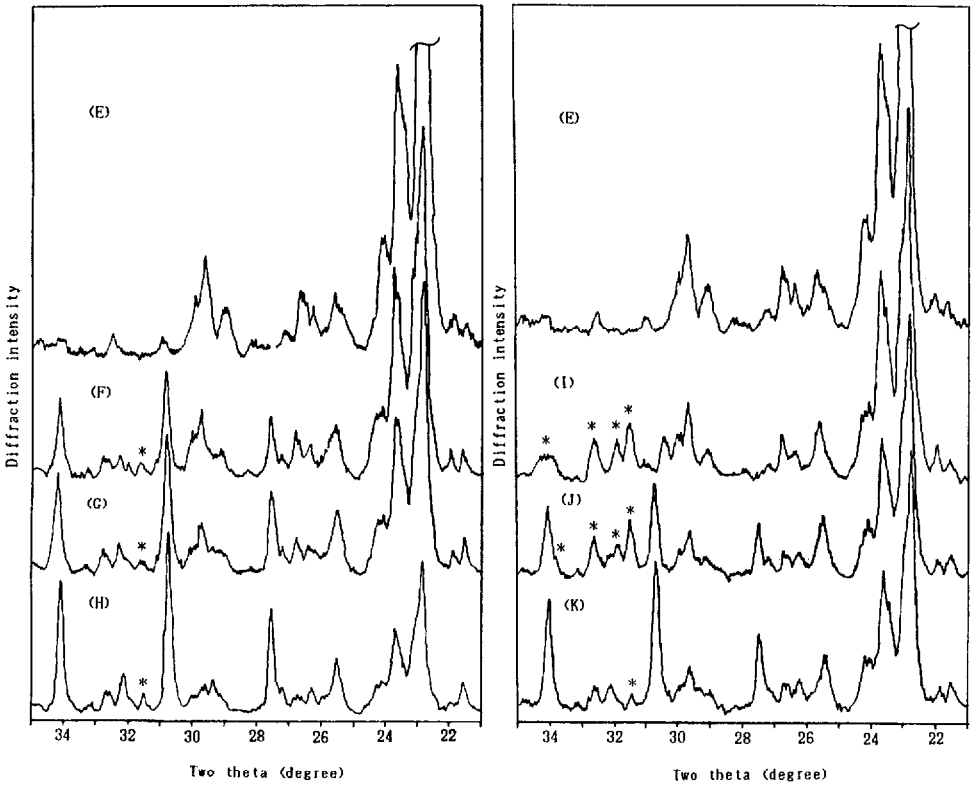

Fig. 3. X-ray diffraction patterns of phosphorus-modified HZSM-5 calcined at 900°C; (E) HZSM-5; (F) HZSM-5/α-TCP(20%)/HPO_4^{2-}; (G) HZSM-5/α-TCP(30%)/HPO_4^{2-}; (H) HZSM-5/α-TCP(50%)/HPO_4^{2-}; (I) HZSM-5/α-TCP(20%)/OH^-; (J) HZSM-5/α-TCP(34%)/OH^-; (K) HZSM-5/α-TCP(34%)/H_2O; (*) hydroxyapatite.

in the particular case with 20% α-TCP it is converted into stoichiometric hydroxyapatite (Fig. 3I).

With 34% α-TCP, neutral hydrothermal treatment gives NHA completely (Fig. 3K) with $x=1$. Here the hydrothermal treatment is probably influenced by the presence of HZSM-5 solid acid.

Methanol conversion

Various views have been expressed on the mechanism of the conversion of methanol to hydrocarbons over HZSM-5 [18,19]. Although there is ambiguity about the detailed mechanism of the initial formation of ethylene, it is plausible that higher olefins are formed by the methylation and oligomerization of ethylene over acid HZSM-5 catalyst. At a high temperature of methanol conversion, cracking of the higher olefins also occurs to give ethylene and propylene in high yields. At strong acid sites, oligomerization and cyclization of olefins occur to give aromatic compounds, and subsequent hydrogen transfer forms alkanes and fused-ring aromatic compounds but in modest amounts. The fused-ring aromatic compounds are said to the precursors of coking and hence their formation results in catalyst deactivation. Possibly detection of BTX is indicative of coke generation on the catalyst. It is generally assumed that while coke deposition proceeds, methanol conversion is maintained at nearly 100%, but higher olefins and alkanes increase because of a decrease in cracking activity, and therefore light olefins decrease whereas BTX does not change much.

The last column in Table 1 shows the catalyst lifetimes at 600°C. This is the time during which methanol conversion is ca. 100%, with no residual dimethyl ether and with a selectivity for ethylene+propylene (C_2+C_3) of >50%. In

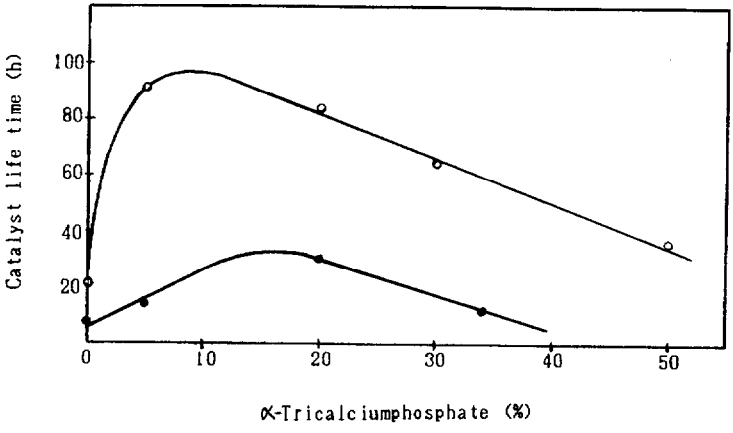

Fig. 4. Relationships between catalyst lifetime and amount of α-TCP added (○) HZSM-5/α-TCP/HPO_4^{2-}; (●) HZSM-5/α-TCP.

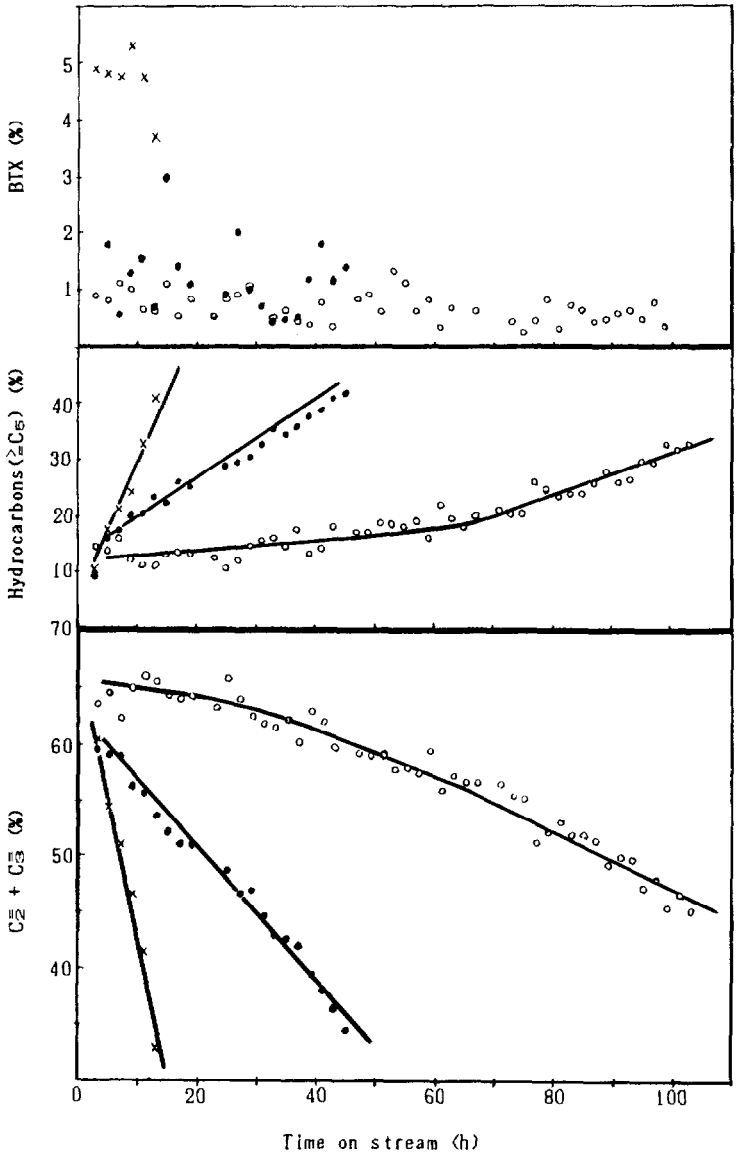

Fig. 5. Selectivities of hydrocarbons vs. time on-stream: (○) HZSM-5/α-TCP(5%)/HPO_4^{2-}; (●) HZSM-5/HPO_4^{2-}; (×) HZSM-5.

Fig. 4 the lifetime is plotted against amount of α-TCP added for both the catalysts hydrothermally treated under slightly acidic conditions and the catalysts without treatment. The results indicates that the maximum catalyst life is achieved with the addition of ca. 10% of α-TCP for the treated catalyst

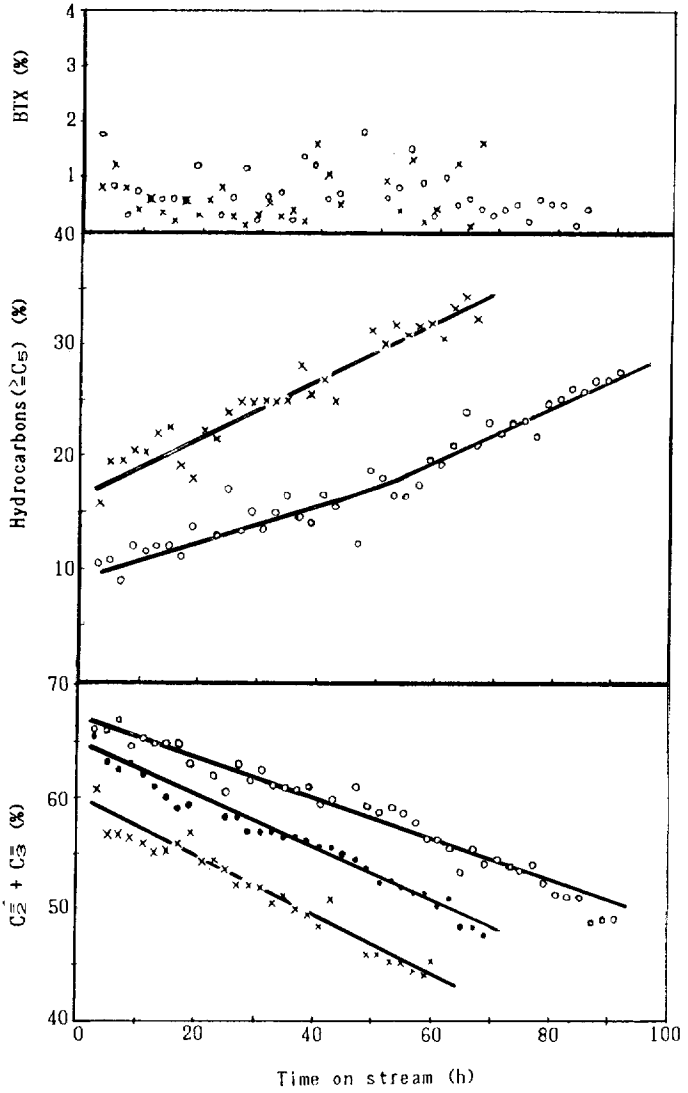

Fig. 6. Selectivities of hydrocarbons vs. time on-stream: (○) HZSM-5/α-TCP(20%)/HPO_4^{2-}; (●) HZSM-5/α-TCP(30%)/HPO_4^{2-}; (×) HZSM-5/α-TCP(50%)/HPO_4^{2-}.

system, but slightly more for the untreated catalyst. The lifetime of catalyst 2P (5% α-TCP addition) was 4.5 times longer than that of HZSM-5 treated with HPO_4^{2-} (catalyst 1P) and more than 10 times longer than that of the parent HZSM-5 (catalyst 1). Even without the HPO_4^{2-} treatment, the lifetime of catalyst 3 was 4 times that of the parent HZSM-5 with addition of α-TCP.

TABLE 2

Methanol conversion with various calcium phosphates

Calcium phosphate	Catalysis[a]							
	Methanol conversion (%)	Yield (%)						
		DME[b]	CO	CO_2	CH_4	$C_2^=+C_3^=$	BTX	Hydrocarbons[c]
$Ca_2P_2O_7$	25.9	13.7	0.1	0.1	0.4	0.2	0.1	10.5
β-TCP	44.4	25.0	2.3	0.1	3.2	0.4	0.0	11.5
NHA[d]	33.1	15.0	0.6	0.0	1.6	0.2	0.0	15.9
Hydroxyapatite	88.8	0.9	58	9.1	7.8	0.9	0.1	21.6

[a]Reaction temperature, 600°C; time, 3 h; WHSV, 7.0 h^{-1}.
[b]Dimethyl ether.
[c]Hydrocarbons with carbon number greater than 4.
[d]$Ca_{10-x}(HPO_4)_{6-x}(OH)_{2-x}$, obtained by the slightly acidic treatment of α-TCP.

TABLE 3

Adsorption properties of hexane isomers for HZSM-5 and phosphorus-modified zeolites

Catalyst	Sample weight (mg)	Hexane isomers (pulse numbers)		
		2,2-Dimethylbutane	3-Methylpentane	n-Hexane
1	100	1	8	21
2P	120[a]	1	8	21
3P	120[a]	1	8	21
NHA[b]	100	1	1	1

[a]Weight of HZSM-5 in the sample, 100 mg.
[b]$Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$.

In order to understand the effects of the HPO_4^{2-} treatment and the presence of α-TCP (i.e., NHA with composition $Ca_3(PO_4)_2$ after hydrothermal treatment) on the catalytic behaviour of HZSM-5, the selectivities for C_2+C_3, BTX and $C_{\geqslant 5}$ hydrocarbons are plotted against time on-stream in Fig. 5. The catalyst lifetime with regard to C_2+C_3 selectivity is very much improved by the HPO_4^{2-} treatment and the presence of NHA [See ▲ in Fig. 7 for the HZSM-5/α-TCP (5%) catalyst].

The hydrogen form of ZSM-5 became inactive after a short reaction time mainly because of coking, as the relatively high BTX selectivity and the increase in the selectivity for $C_{\geqslant 5}$ hydrocarbons suggest, but also because of dealumination of the framework aluminium discussed later. Deactivation due to coke deposition was depressed by the HPO_4^{2-} treatment of HZSM-5 and almost prevented by the presence of NHA. The NHA in HZSM-5, essentially

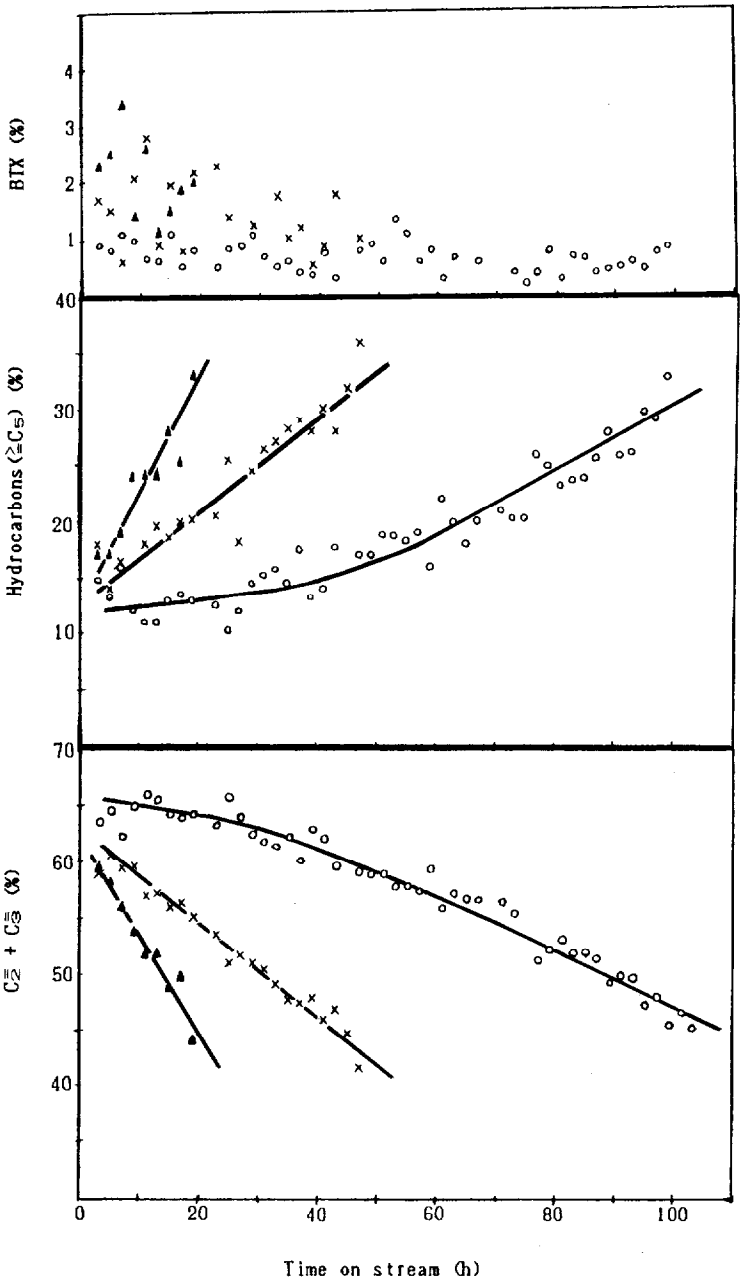

Fig. 7. Selectivities of hydrocarbons vs. time on-stream: (▲) HZSM-5/α-TCP(5%); (○) HZSM-5/α-TCP)5%)/HPO_4^{2-}; (×) HZSM-5/α-TCP(5%)/OH^-.

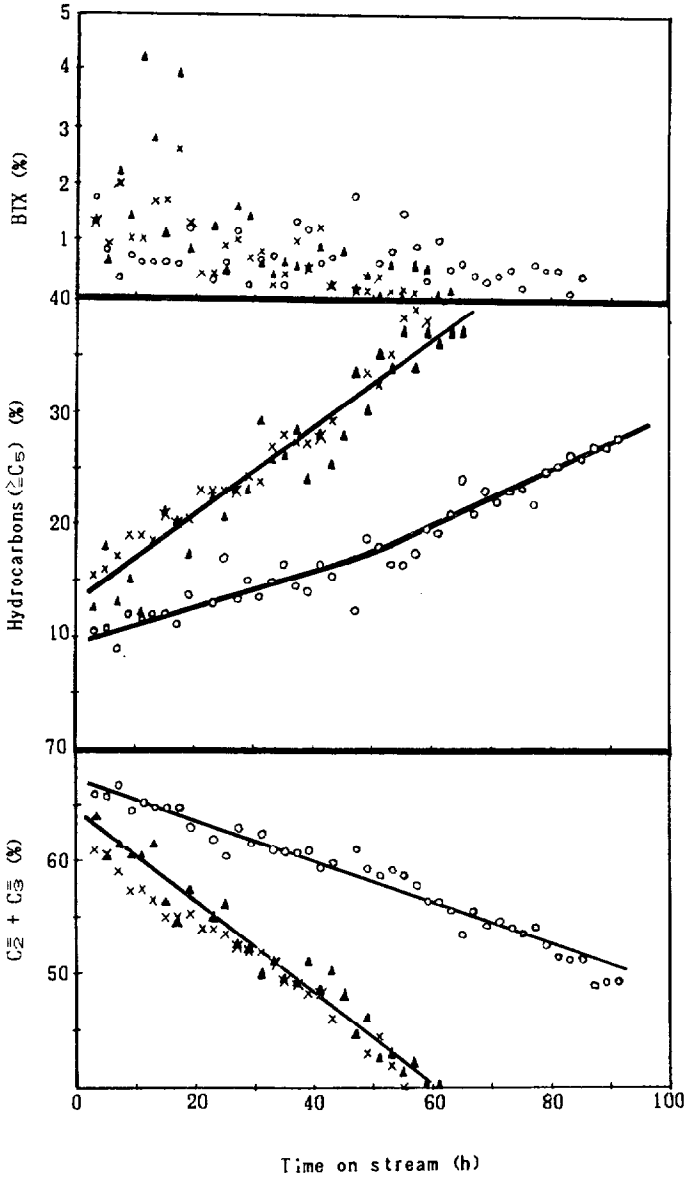

Fig. 8. Selectivities of hydrocarbons vs. time on-stream: (○) HZSM-5/α-TCP(20%)/HPO_4^{2-}; (▲) HZSM-5/α-TCP(20%); (×) HZSM-5/α-TCP(20%)/OH^-.

located on the external surface of HZSM-5 crystals, and the HPO_4^{2-} moieties in the HZSM-5 crystals seem to prohibit coke formation.

As shown in Fig. 4, ca. 10% α-TCP addition is necessary for the greatest extension of the catalyst lifetime, but a larger amount of NHA is undesirable.

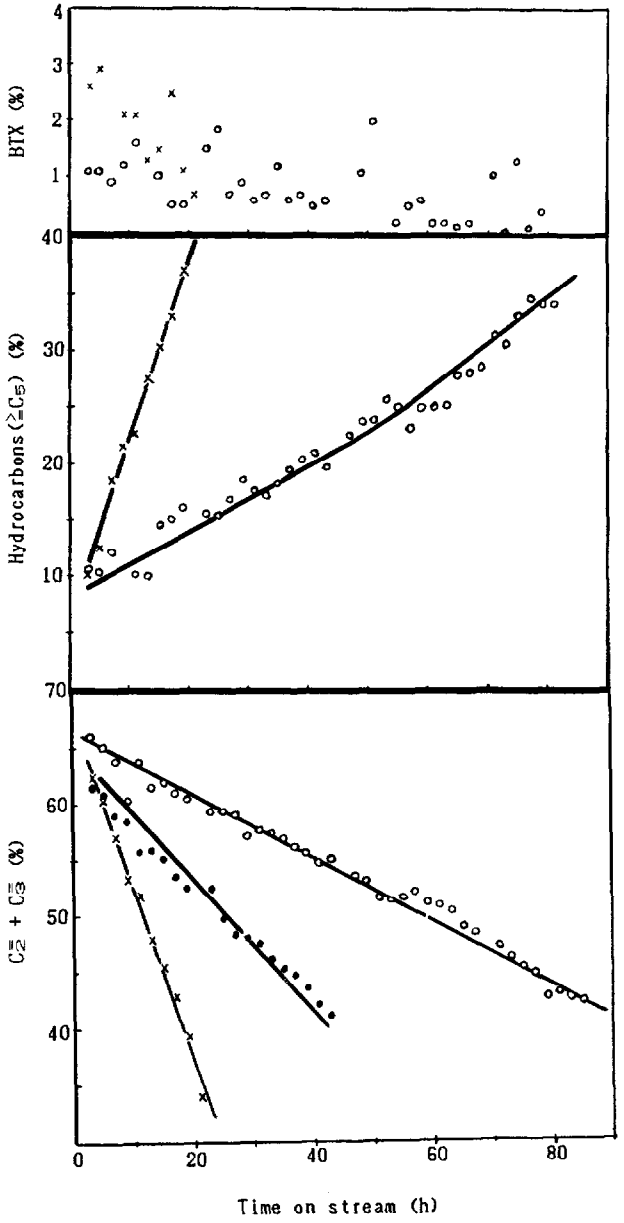

Fig. 9. Selectivities of hydrocarbons vs. time on-stream: (×) HZSM-5/α-TCP(34%); (●) HZSM-5/α-TCP(34%)/H_2O; (○) HZSM-5/α-TCP (34%)/OH^-.

Fig. 6 shows the selectivities for C_2+C_3 of catalysts 3P, 4P and 6P and those for $C_{\geqslant 5}$ hydrocarbons and BTX for catalysts 3P and 6P. When NHA is present in a larger amount, the selectivities for C_2+C_3 are lower whereas those of $C_{\geqslant 5}$ hydrocarbons are higher. On the other hand, the degree of BTX formation does not appear to change.

These results indicate that the rate of deactivation by coking does not depend on the amount of NHA. The lower selectivities for C_2+C_3 with larger amounts of NHA seem to correspond to the higher values for $C_{\geqslant 5}$ hydrocarbons. NHA itself produces $C_{\geqslant 5}$ hydrocarbons to some extent in methanol conversion at 600 °C, as shown in Table 2, so that excess of NHA is considered to cause the decrease in C_2+C_3 selectivity shown in Figs. 4 and 6. The minimum amount of α-TCP (NHA in practice) required for effective modification may be less than 5 wt.-% with respect to the parent HZSM-5. The roles of the modifiers are described in detail later.

In Figs. 7, 8 and 9, the selectivities for C_2+C_3 and BTX are plotted against time on-stream in order to compare the influences of different hydrothermal treatments on catalyst performance for catalysts 2, 3 and 5. Basic treatment (catalysts 2B and 5B) improves the catalyst lifetime because of more moderate coke generation, as indicated by the lower selectivities for BTX and $C_{\geqslant 5}$ hydrocarbons.

On basic treatment, α-TCP added to HZSM-5 is converted into NHA ($x \ll 1$), releasing HPO_4^{2-} into aqueous solution, which modifies HZSM-5. This seems to be the reason for the extended lifetime of catalysts 2B and 5B. This speculation is supported by the result that the lifetime of catalysts 5B (64 h) is close to that resulting from addition of 34% α-TCP (curve o) in Fig. 4). The improvement in catalyst lifetime by the neutral hydrothermal treatment (catalyst 5A) was less than that given by the basic treatment (catalyst 5B), corresponding with a smaller release of HPO_4^{2-} by the conversion of α-TCP into NHA (with higher x) by neutral treatment.

The effects of both HPO_4^{2-} and α-TCP on the catalyst lifetime were similar even when a slightly acidic hydrothermal treatment was performed on HZSM-5 before pelletization with α-TCP (catalyst 5P).

One of the ways in which catalyst 3B differs from catalysts 2B and 5B is that the α-TCP is completely hydrolysed to stoichiometric hydroxyapatite. As shown in Table 2, hydroxyapatite catalyses methanol conversion to $C_{\geqslant 5}$ hydrocarbons in high yield, but in fact the $C_{\geqslant 5}$ hydrocarbon selectivity of catalyst 3 did not decrease after the basic treatment though the BTX selectivity decreased slightly.

Roles of calcium phosphates and HPO_4^{2-}

With a slightly acidic hydrothermal treatment, α-TCP on HZSM-5 was converted exclusively into NHA, which forms as tiny agglomerates of thin blades

in hardened pellets with parent HZSM-5. HZSM-5 was also modified with HPO_4^{2-} in the acidic treatment, resulting in superior catalysts.

Previous reports have dealt with the effects of modification of HZSM-5 by impregnating phosphorus compounds [8,9] and boric acid [20] on methanol conversion and toluene alkylation and disproportionation, respectively. Phosphorus and boron as modifiers bond to the zeolite framework through oxygen and occupy acid sites. It is, however, still a subject of argument whether the modifiers lower the acidity of the active sites compared with the parent HZSM-5 (neutralizing acid sites, in particular, strong acid sites on the external surface) or slightly reduce the channel size and increase the tortuosity of the channels of the HZSM-5 crystals. In any event higher yields of light olefins and lower yields of saturated aliphatic and aromatic compounds were obtained as in the present work with α-TCP and HPO_4^{2-}.

Though modification of HZSM-5 solely with HPO_4^{2-} achieved a moderate improvement of the catalyst lifetime, the latter was greatly prolonged when further combined with grinding together of HZSM-5 with α-TCP. The question then arises of what the roles of α-TCP and/or NHA in methanol conversion are.

The observed rapid deactivation of HZSM-5 is assigned to coke deposition and, in particular, coke on the external surface completely deactivates ca. 0.4 μm HZSM-5. Therefore, reducing acid sites on the external surface before methanol conversion may result in a longer lifetime of HZSM-5.

At first we considered that in grinding together HZSM-5 and α-TCP, acid sites on the external surface are reduced according to the equation

$$\alpha\text{-}Ca_3(PO_4)_2 + 2H^+ \longrightarrow 2CaHPO_4 + Ca^{2+}$$

If this reaction actually occurs, $Ca_2P_2O_7$, which is formed by calcination of $CaHPO_4$ at 500°C, should be identified by XRD analysis of the zeolites ground together with α-TCP. In fact, $Ca_2P_2O_7$ was not observed because of the extremely low external acid concentration. We then examined the conversions of α-TCP in acetic acid solution, and modification of HY-type zeolite (Si/Al=2.6) by grinding with 20% α-TCP for a longer time (5 h). Two samples were analysed by XRD after calcination at 900°C, as shown in Fig. 10. The formation of appreciable amounts of β-$Ca_2P_2O_7$ and β-TCP was observed for the sample obtained from acetic acid solution. For the modified HY zeolite, a low-intensity peak (marked with an asterisk) was observed as a shoulder on the HY zeolite peak (Fig. 10M) and is assigned to β-$Ca_2P_2O_7$.

In the catalyst system HZSM-5/α-TCP/HPO_4^{2-}, coke deposition on the external surface was reduced, which might make it possible to diffuse reactants and products into or from the HZSM-5 channels for a longer conversion time. The efficiency of the solid–solid reaction of α-TCP with acid sites on the external surface seems to be lower with the addition of 5% α-TCP (catalyst 2) than that in other instances where the amount of α-TCP is more than 20%,

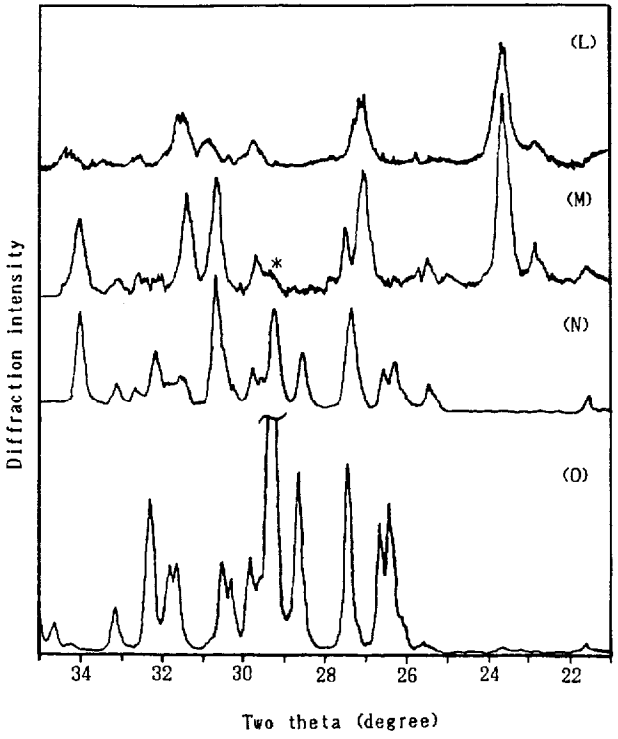

Fig. 10. X-ray diffraction patterns after calcination at 900°C: (L) HY zeolite (Si/Al-2.6); (M) HY zeolite/α-TCP(15%); (N) α-TCP/CH_3COOH; (O) β-$Ca_2P_2O_7$; (*) β-$Ca_2P_2O_7$.

although the decrease in C_2+C_3 selectivities by NHA formed from α-TCP is not negligible. This would explain the maximum catalyst lifetime between 5 and 20% of α-TCP.

The dotted lines in Fig. 11 indicate the C_2+C_3 selectivities measured immediately after each regeneration (550°C in air). The slope of the line for the parent HZSM-5 is sharp, indicating that the dealumination of HZSM-5 in by-product water occurred intensely more than that for the modified HZSM-5 (Catalyst 2P). HPO_4^{2-} is considered not only to modify acid sites but also to retard dealumination.

It was mentioned before that calcium phosphates, either NHA or a stoichiometric hydroxyapatite, is undesirable on account of their catalytic activities to give $C_{\geqslant 5}$ hydrocarbons in higher yields. However, NHA and a stoichiometric hydroxyapatite on HZSM-5 can still differ in their effect on catalyst performance, provided that NHA gradually releases HPO_4^{2-} to HZSM-5 and is converted into a stoichiometric hydroxyapatite in the presence of water from methanol conversion at high temperature:

$$Ca_{10-x}(HPO_4)_x(PO_4)_{6-x}(OH)_{2-x}$$
$$+nH_2O \longrightarrow Ca_{10-y}(HPO_4)_y(PO_4)_{6-y}(OH)_{2-y}+(x-y)HPO_4^{2-} \quad (x>y)$$

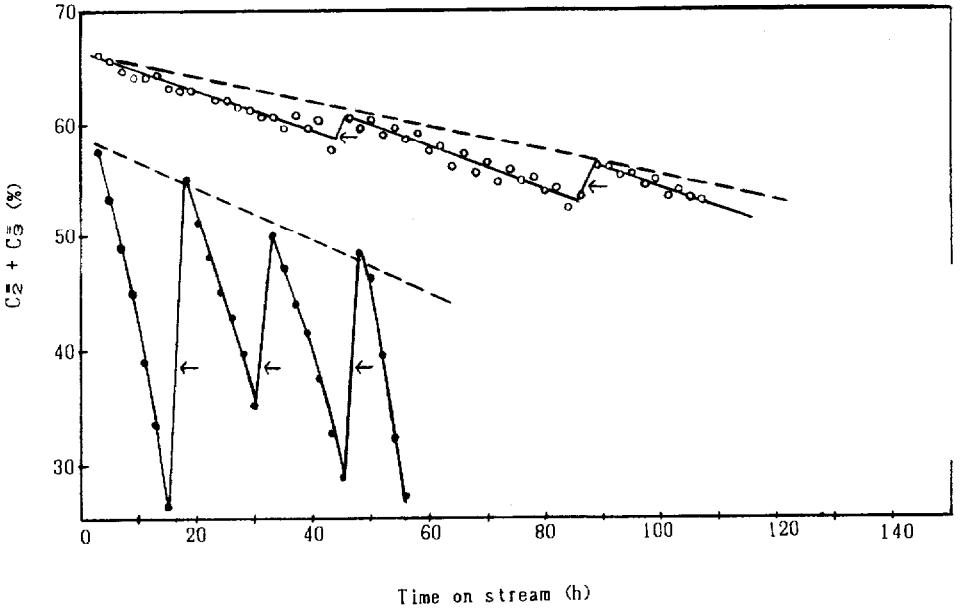

Fig. 11. Selectivities of $C_2^= + C_3^=$ vs. time on-stream including regenerations (←) in air at 550°C for 6 h: (○) catalyst 2P; (●) catalyst 1.

As reported for boron-modified HZSM-5 [20], boric acid mobilizes in HZSM-5 crystals in the presence of water and the mobility of the boric acid is a factor in determining the catalytic activity and selectivity in toluene alkylation by methanol and disproportionation. Hence it is expected that HPO_4^{2-} released from NHA restricts the mobility of phosphorus moieties which play an important role in reducing coke deposition and/or dealumination.

Therefore, in order to confirm the conversion of NHA into NHA with x reduced almost to zero in methanol conversion, XRD analysis was performed on catalyst 3P used for the catalyst lifetime test and calcined at 900°C (Fig. 12). The analysis also included catalyst 3P steamed at 600°C for 5 days. Obviously the peaks assigned to hydroxyapatite (marked with asterisks) showed a higher intensity for catalyst 3P used in the methanol conversion and were much higher for catalyst 3P steamed under very severe conditions.

The adsorption properties of hexane isomers for HZSM-5, HZSM-5/α-TCP (20%), HZSM-5/α-TCP (20%)/HPO_4^{2-} were examined to establish whether phosphorus compounds decrease the pore size and/or the pore volume of HZSM-5. The pulse numbers are considered to be proportional to the pore volume for the same zeolite weight and the same hexane isomer, and the ratios of pulse numbers among hexane isomers depend on the pore size. As a result,

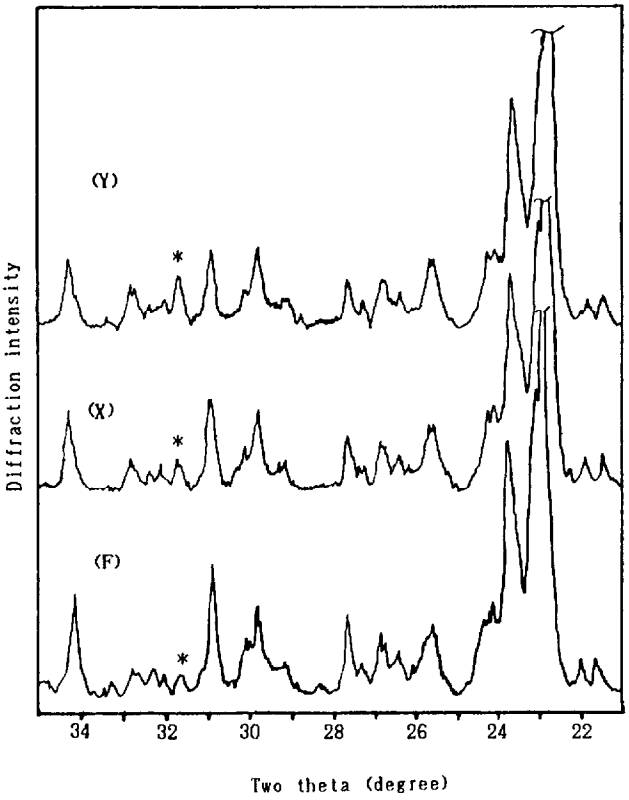

Fig. 12. X-ray diffraction patterns of phosphorus-modified HZSM-5 calcined at 900°C: (F)p HZSM-5/α-TCP(20%)/HPO_4^{2-}; (X) used for a catalyst lifetime test at 600°C; (Y) steamed for 5 days at 600°C.

in our experiment the pulse numbers and the ratios are similar among the catalysts.

Finally, it seems reasonable to conclude that in the catalyst system HZSM-5/α-TCP/HPO_4^{2-}, α-TCP on HZSM-5 is converted into NHA on hydrothermal treatment and it gradually supplies HPO_4^{2-} to HZSM-5, resulting in longer catalyst lifetime. In the HZSM-5/α-TCP system, deactivation occurs rapidly before HPO_4^{2-} from NHA modifies the acid sites.

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